

Decision Trees and Tree-Based ML Algorithms

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Learning Objectives

After completing this lecture, students will be able to:

- Explain the motivation behind tree-based learning.
- Describe the structure of a Decision Tree.
- Identify root nodes, internal nodes, branches, and leaf nodes.
- Differentiate between Classification Trees and Regression Trees.
- Explain the strengths and weaknesses of Decision Trees.
- Understand why a splitting criterion is required.

Why Tree-Based Models?

Consider the following problem:

Loan Approval

A bank wants to decide whether a loan application should be approved.

Decision may depend on:

- Age
- Income
- Employment Status
- Credit Score

Question:

Can a simple linear equation capture all these decision rules?

Often, the answer is **No**.

Why Linear Models Sometimes Fail

Linear models assume:

$$y = wx + b$$

They work well when relationships are approximately linear.

However, many real-world problems involve:

- Complex decision rules
- Multiple conditions
- Non-linear boundaries

Examples:

- Loan Approval
- Disease Diagnosis
- Customer Churn Prediction
- Fraud Detection

Decision Trees handle such situations naturally.

How Humans Make Decisions

Suppose a student asks:

Should I Apply for a Scholarship?

A teacher may think:

- ① Is CGPA greater than 3.50?
- ② If Yes:
 - Apply
- ③ If No:
 - Improve academic results first

Humans naturally make decisions using a sequence of IF-THEN rules.
Decision Trees mimic this process.

What is a Decision Tree?

Definition

A Decision Tree is a supervised learning algorithm that recursively splits data into smaller subsets using decision rules.

Characteristics:

- Easy to understand
- Rule-based model
- Works for classification
- Works for regression
- Produces interpretable results

A Simple Decision Tree

Age > 30 ?

Yes → Buy Product

No

Income > 50K ?

Yes → Buy Product

No → Do Not Buy

Notice:

- Multiple decisions are made sequentially.
- Each decision reduces uncertainty.

Components of a Decision Tree

Root Node

The first decision point of the tree.

Internal Node

Intermediate decision points.

Branch

Connection between nodes.

Leaf Node

Final prediction or outcome.

Tree Terminology Example

Example:

Root Node

$Age > 30$

- Root Node = $Age > 30$
- Branch = Yes / No
- Internal Node = $Income > 50K$
- Leaf Node = Buy or Not Buy

Depth of Tree

- Number of decisions from the root node to a leaf node.

Classification Tree

Classification Trees predict categories.

Example:

Study Hours > 5

- Yes $\rightarrow$ Pass
- No $\rightarrow$ Fail

Output:

- Pass
- Fail

These are discrete categories.

Classification Problems

Common applications:

- Email Spam Detection
- Disease Diagnosis
- Loan Approval
- Customer Churn Prediction
- Sentiment Analysis

Output is always a class label.

Examples:

- Yes / No
- Positive / Negative
- Approved / Rejected

Regression Tree

Regression Trees predict continuous values.

Example:

Area > 1500 sq.ft

- Yes $\rightarrow$ Price = \$250,000
- No $\rightarrow$ Price = \$150,000

Output:

- Numerical value
- Continuous prediction

Classification vs Regression Trees

Aspect	Classification	Regression
Output	Category	Numeric Value
Example	Pass/Fail	House Price
Target Type	Discrete	Continuous

Advantages of Decision Trees

- Easy to understand and explain.
- Produces human-readable rules.
- Works with numerical and categorical data.
- No feature scaling required.
- Handles non-linear relationships.
- Fast prediction.

Limitations of Decision Trees

- Can overfit training data.
- Sensitive to small changes in data.
- May create very complex trees.
- High variance model.
- Greedy learning process.

Question:

How does a Decision Tree decide which feature should be used first?

Need for a Splitting Criterion

Suppose we have three features:

- Age
- Income
- Education

Question:

Which feature should become the root node?

Age?
Income?
Education?

We need a mathematical way to determine the best split.

In the next lecture, we will learn:

- Entropy
- Information Gain
- Gini Index

Entropy Concept

Question:

How impure or uncertain is a dataset?

Entropy measures the amount of uncertainty in a dataset.

Examples:

- All samples belong to one class

$$Entropy = 0$$

- Classes are evenly mixed

$$Entropy = 1$$

Higher entropy means higher uncertainty.

Understanding Entropy Intuitively

Case 1:

[Yes, Yes, Yes, Yes]

Can we predict the class?

Absolutely Yes

Entropy:

0

Case 2:

[Yes, No, Yes, No]

Prediction becomes difficult.

Entropy becomes high.

Entropy Formula

Entropy is defined as:

$$Entropy(S) = - \sum_{i=1}^c p_i \log_2(p_i)$$

Where:

- p_i = Probability of class i
- c = Number of classes

Goal:

- Lower Entropy = Better Split
- Higher Entropy = More Uncertainty

Entropy Example Dataset

Consider the following training dataset:

ID	Age	Buy Laptop
1	Young	No
2	Young	No
3	Young	Yes
4	Old	Yes
5	Old	Yes
6	Old	Yes

Target Variable:

Buy Laptop

Classes:

- Yes = 4
- No = 2

Calculating Class Probabilities

From the dataset:

- Total Samples = 6
- Yes = 4
- No = 2

Probability of Yes:

$$P(\text{Yes}) = \frac{4}{6} = 0.667$$

Probability of No:

$$P(\text{No}) = \frac{2}{6} = 0.333$$

These probabilities will be used to calculate Entropy.

Entropy Calculation

Entropy Formula:

$$Entropy(S) = - \sum p_i \log_2(p_i)$$

Substituting probabilities:

$$Entropy(S) = - (0.667 \log_2(0.667) + 0.333 \log_2(0.333))$$

$$= - (0.667(-0.585) + 0.333(-1.585))$$

$$= 0.918$$

$$Entropy(S) = 0.918$$

Entropy Interpretation

Class Distribution	Entropy
100% Yes	0
80% Yes, 20% No	0.722
67% Yes, 33% No	0.918
50% Yes, 50% No	1

Observation:

- Entropy increases as uncertainty increases.
- Maximum entropy occurs when classes are equally distributed.

Question:

Which feature should be chosen as the root node?

Decision Trees use Information Gain.

Definition:

$$IG = Entropy(Parent) - WeightedEntropy(Children)$$

Goal:

Maximize Information Gain

Higher Information Gain means a better split.

Young Group After Split

Original Dataset:

ID	Age	Income	Buy Laptop
1	Young	High	No
2	Young	Low	No
3	Young	High	Yes
4	Old	High	Yes
5	Old	Low	Yes
6	Old	Low	Yes

Highlighted rows belong to the Young group.

Splitting the Dataset by Age

Young Group:

[No, No, Yes]

Old Group:

[Yes, Yes, Yes]

Now calculate the entropy of each group.

Entropy of Young Group

Young Group:

[No, No, Yes]

Probabilities:

$$P(\text{No}) = \frac{2}{3}$$

$$P(\text{Yes}) = \frac{1}{3}$$

Entropy:

$$\begin{aligned} \text{Entropy}(\text{Young}) &= - \left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3} \right) \\ &= 0.918 \end{aligned}$$

Entropy of Old Group

Old Group:

[Yes, Yes, Yes]

Probabilities:

$$P(\text{Yes}) = 1$$

Entropy:

$$\begin{aligned} \text{Entropy}(\text{Old}) &= -(1) \log_2(1) \\ &= 0 \end{aligned}$$

Perfectly pure node.

Weighted Entropy

Weighted Entropy:

$$= \frac{3}{6}(0.918) + \frac{3}{6}(0)$$

$$= 0.459$$

$$\boxed{\textit{WeightedEntropy} = 0.459}$$

Information Gain Calculation

Parent Entropy:

0.918

Weighted Child Entropy:

0.459

Information Gain:

$$IG = 0.918 - 0.459$$

$$= 0.459$$

$$IG = 0.459$$

Interpretation of Information Gain

Before splitting:

$$Entropy = 0.918$$

After splitting:

$$Entropy = 0.459$$

Reduction:

$$0.918 - 0.459 = 0.459$$

Information Gain measures how much uncertainty is removed by a split.
Higher Information Gain means better splitting.

Another impurity measure:

$$Gini(S) = 1 - \sum p_i^2$$

Used by:

- Scikit-Learn DecisionTreeClassifier
- Random Forest

Smaller Gini values indicate purer nodes.

Gini Calculation

Dataset:

$$P(Yes) = 0.667$$

$$P(No) = 0.333$$

Formula:

$$Gini = 1 - (0.667^2 + 0.333^2)$$

$$= 1 - (0.444 + 0.111)$$

$$= 0.445$$

$Gini = 0.445$

Gini Interpretation

Distribution	Gini
100% One Class	0
80% / 20%	0.32
67% / 33%	0.445
50% / 50%	0.50

Lower Gini indicates lower impurity.

Entropy vs Gini

Feature	Entropy	Gini
Formula	Logarithmic	Simple
Computation	Slower	Faster
Interpretability	Higher	Moderate
Used By	ID3, C4.5	CART, Random Forest

In practice, both often produce similar trees.

Key Takeaways

- Entropy measures uncertainty.
- Information Gain measures reduction in uncertainty.
- The best split maximizes Information Gain.
- Gini Index measures impurity.
- Lower Gini indicates a better split.
- Decision Trees use these measures to build trees automatically.

Building a Decision Tree Step-by-Step

Recall our dataset:

ID	Age	Income	Buy Laptop
1	Young	High	No
2	Young	Low	No
3	Young	High	Yes
4	Old	High	Yes
5	Old	Low	Yes
6	Old	Low	Yes

Goal:

Find the feature that best separates the classes.

Step 1: Calculate Parent Entropy

Target variable:

Buy Laptop

Class distribution:

Yes = 4, *No* = 2

Probabilities:

$$P(\text{Yes}) = \frac{4}{6} = 0.667$$

$$P(\text{No}) = \frac{2}{6} = 0.333$$

Parent Entropy:

$$Entropy(S) = - (0.667 \log_2(0.667) + 0.333 \log_2(0.333))$$

$$Entropy(S) = 0.918$$

Step 2: Candidate Features

We have two input features:

Age

Income

To build the tree, we calculate:

$IG(Age)$

and

$IG(Income)$

Then we choose the feature with the highest Information Gain.

Step 3: Split by Age

Split the dataset by Age:

Age	IDs	Class Distribution	Entropy
Young	1,2,3	1 Yes, 2 No	?
Old	4,5,6	3 Yes, 0 No	?

Now calculate entropy for each group.

Step 4: Entropy of Young Group

Young group:

ID	Age	Income	Buy Laptop
1	Young	High	No
2	Young	Low	No
3	Young	High	Yes

Class distribution:

$$\text{Yes} = 1, \quad \text{No} = 2$$

$$\text{Entropy}(\text{Young}) = - \left(\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3} \right)$$

$$\boxed{\text{Entropy}(\text{Young}) = 0.918}$$

Step 5: Entropy of Old Group

Old group:

ID	Age	Income	Buy Laptop
4	Old	High	Yes
5	Old	Low	Yes
6	Old	Low	Yes

Class distribution:

$$Yes = 3, \quad No = 0$$

Since all samples belong to one class:

$$Entropy(Old) = 0$$

This is a pure node.

Step 6: Information Gain for Age

Weighted Entropy after splitting by Age:

$$WE(Age) = \frac{3}{6}(0.918) + \frac{3}{6}(0)$$

$$WE(Age) = 0.459$$

Information Gain:

$$IG(Age) = Entropy(S) - WE(Age)$$

$$IG(Age) = 0.918 - 0.459$$

$$IG(Age) = 0.459$$

Step 7: Split by Income

Split the dataset by Income:

Income	IDs	Class Distribution	Entropy
High	1,3,4	2 Yes, 1 No	?
Low	2,5,6	2 Yes, 1 No	?

Now calculate entropy for each Income group.

Step 8: Entropy of High Income Group

High Income group:

ID	Age	Income	Buy Laptop
1	Young	High	No
3	Young	High	Yes
4	Old	High	Yes

Class distribution:

$$\text{Yes} = 2, \quad \text{No} = 1$$

$$\text{Entropy}(\text{High}) = - \left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3} \right)$$

$$\boxed{\text{Entropy}(\text{High}) = 0.918}$$

Step 9: Entropy of Low Income Group

Low Income group:

ID	Age	Income	Buy Laptop
2	Young	Low	No
5	Old	Low	Yes
6	Old	Low	Yes

Class distribution:

$$\text{Yes} = 2, \quad \text{No} = 1$$

$$\text{Entropy}(\text{Low}) = - \left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3} \right)$$

$$\boxed{\text{Entropy}(\text{Low}) = 0.918}$$

Step 10: Information Gain for Income

Weighted Entropy after splitting by Income:

$$WE(Income) = \frac{3}{6}(0.918) + \frac{3}{6}(0.918)$$

$$WE(Income) = 0.918$$

Information Gain:

$$IG(Income) = Entropy(S) - WE(Income)$$

$$IG(Income) = 0.918 - 0.918$$

$$IG(Income) = 0$$

Step 11: Choose the Root Node

Information Gain comparison:

$$IG(Age) = 0.459$$

$$IG(Income) = 0$$

Since:

$$IG(Age) > IG(Income)$$

The best feature for the root node is:

Age

Step 12: First Split of the Tree

Root Node:

Age

Condition	Class Distribution	Decision
Age = Old	3 Yes, 0 No	Buy Laptop = Yes
Age = Young	1 Yes, 2 No	Need Further Split

The Old branch becomes a leaf node.

The Young branch still has mixed classes.

Step 13: Analyze Young Subset

Young subset:

ID	Age	Income	Buy Laptop
1	Young	High	No
2	Young	Low	No
3	Young	High	Yes

Remaining feature:

Income

Now split the Young subset by Income.

Step 14: Split Young Subset by Income

Young subset after splitting by Income:

Income	IDs	Class Distribution	Decision
Low	2	0 Yes, 1 No	No
High	1,3	1 Yes, 1 No	Mixed

For Income = Low:

$$\text{BuyLaptop} = \text{No}$$

For Income = High:

$$\text{Yes} = 1, \quad \text{No} = 1$$

This branch is still impure.

Step 15: What Happens to the Mixed Branch?

For the branch:

$$\text{Age} = \text{Young}, \quad \text{Income} = \text{High}$$

We have:

ID	Age	Income	Buy Laptop
1	Young	High	No
3	Young	High	Yes

Both samples have the same feature values but different target values.
Therefore, no remaining feature can separate them.

A common solution is majority voting.

Here:

$$\text{Yes} = 1, \quad \text{No} = 1$$

Tie can be resolved by using parent majority class.

Parent Young subset majority:

$$\text{No} = 2, \quad \text{Yes} = 1$$

So prediction:

$$\text{BuyLaptop} = \text{No}$$

Final Decision Tree

The final learned rules are:

If Age = Old, then Buy Laptop = Yes

If Age = Young and Income = Low, then Buy Laptop = No

If Age = Young and Income = High, then Buy Laptop = No

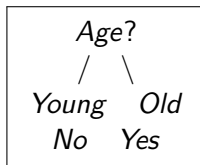
So the simplified tree becomes:

Age = Old $\rightarrow$ Yes

Age = Young $\rightarrow$ No

Final Tree Interpretation

Final Decision Tree:



Interpretation:

- Age is the most important feature.
- Income does not provide useful Information Gain at the root.
- Old customers always buy laptop in this dataset.
- Young customers mostly do not buy laptop.

Another Full Decision Tree Example Dataset

ID	Age	Income	Student	CreditScore	Buy Laptop
1	Young	High	No	650	No
2	Young	High	No	750	No
3	Middle	High	No	700	Yes
4	Senior	Medium	No	680	Yes
5	Senior	Low	Yes	620	Yes
6	Senior	Low	Yes	800	No
7	Middle	Low	Yes	790	Yes
8	Young	Medium	No	660	No
9	Young	Low	Yes	640	Yes
10	Senior	Medium	Yes	710	Yes
11	Young	Medium	Yes	730	Yes
12	Middle	Medium	No	780	Yes
13	Middle	High	Yes	670	Yes
14	Senior	Medium	No	760	No

Features: Age, Income, Student, CreditScore, Target: Buy Laptop

Step 1: Parent Entropy

Total samples:

14

Class distribution:

Yes = 9, *No* = 5

Probabilities:

$$P(\textit{Yes}) = \frac{9}{14} = 0.643$$

$$P(\textit{No}) = \frac{5}{14} = 0.357$$

Entropy:

$$\textit{Entropy}(S) = - (0.643 \log_2 0.643 + 0.357 \log_2 0.357)$$

Step 2: Information Gain for Age

Split by Age:

Age	Yes	No	Entropy
Young	2	3	0.971
Middle	4	0	0
Senior	3	2	0.971

Weighted Entropy:

$$WE(Age) = \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971)$$

$$WE(Age) = 0.694$$

Information Gain:

$$IG(Age) = 0.940 - 0.694$$

$$IG(Age) = 0.247$$

Step 3: Information Gain for Income

Split by Income:

Income	Yes	No	Entropy
High	2	2	1.000
Medium	4	2	0.918
Low	3	1	0.811

Weighted Entropy:

$$WE(\text{Income}) = \frac{4}{14}(1.000) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811)$$

$$WE(\text{Income}) = 0.911$$

Information Gain:

$$IG(\text{Income}) = 0.940 - 0.911$$

$$IG(\text{Income}) = 0.029$$

Step 4: Information Gain for Student

Split by Student:

Student	Yes	No	Entropy
Yes	6	1	0.592
No	3	4	0.985

Weighted Entropy:

$$WE(Student) = \frac{7}{14}(0.592) + \frac{7}{14}(0.985)$$

$$WE(Student) = 0.788$$

Information Gain:

$$IG(Student) = 0.940 - 0.788$$

$$IG(Student) = 0.152$$

Step 5: Numeric Feature CreditScore

For a numeric feature, Decision Tree tests possible threshold values.

Example split:

$$CreditScore \leq 735$$

This creates two groups:

$$CreditScore \leq 735$$

and

$$CreditScore > 735$$

For numeric features, the best threshold is selected based on maximum Information Gain.

Step 6: CreditScore Threshold Comparison

Threshold	Weighted Entropy	Information Gain
630	0.893	0.048
645	0.840	0.100
655	0.940	0.000
665	0.915	0.025
675	0.937	0.003
690	0.939	0.001
705	0.924	0.016
720	0.892	0.048
740	0.838	0.102
755	0.915	0.025
770	0.940	0.000
785	0.930	0.010
795	0.827	0.113

Best numeric threshold:

$$\text{CreditScore} \leq 795$$

$$IG(\text{CreditScore}) = 0.113$$

How Do We Choose Thresholds for CreditScore?

For a numeric feature, Decision Tree cannot directly split like categorical features. So, we create candidate thresholds.

$$\text{CreditScore} \leq t$$

and

$$\text{CreditScore} > t$$

Steps:

- 1 Sort all CreditScore values.
- 2 Take the midpoint between two consecutive values.
- 3 Use each midpoint as a candidate threshold.
- 4 Calculate Weighted Entropy and Information Gain.
- 5 Select the threshold with the highest Information Gain.

Sorted CreditScore Values

CreditScore values from the dataset:

620, 640, 650, 660, 670, 680, 700, 710, 730, 750, 760, 780, 790, 800

Candidate thresholds are midpoints.

Example:

$$\frac{620 + 640}{2} = 630$$

$$\frac{640 + 650}{2} = 645$$

$$\frac{790 + 800}{2} = 795$$

So possible thresholds are:

630, 645, 655, ..., 795

Parent Entropy

Before splitting:

$$Yes = 9, \quad No = 5$$

$$P(Yes) = \frac{9}{14}$$

$$P(No) = \frac{5}{14}$$

$$Entropy(S) = - \left(\frac{9}{14} \log_2 \frac{9}{14} + \frac{5}{14} \log_2 \frac{5}{14} \right)$$

$$Entropy(S) = 0.940$$

Threshold Example 1: $CreditScore \leq 630$

Split:

$$CreditScore \leq 630$$

Left Group:

$$ID = 5$$

[Yes]

Right Group:

13 samples

Yes = 8, No = 5

Entropy for Threshold 630

Left Group:

$$Entropy(Left) = 0$$

Right Group:

$$Yes = 8, \quad No = 5$$

$$Entropy(Right) = - \left(\frac{8}{13} \log_2 \frac{8}{13} + \frac{5}{13} \log_2 \frac{5}{13} \right)$$

$$Entropy(Right) = 0.961$$

Weighted Entropy:

$$WE = \frac{1}{14}(0) + \frac{13}{14}(0.961) = 0.893$$

Information Gain for Threshold 630

Parent Entropy:

$$Entropy(S) = 0.940$$

Weighted Entropy:

$$WE = 0.893$$

Information Gain:

$$IG = Entropy(S) - WE$$

$$IG = 0.940 - 0.893$$

$$IG = 0.047$$

So, threshold 630 gives a small reduction in uncertainty.

Threshold Example 2: $CreditScore \leq 795$

Threshold:

$$CreditScore \leq 795$$

Left Group:

$$Yes = 9, \quad No = 4$$

$$Entropy(Left) = - \left(\frac{9}{13} \log_2 \frac{9}{13} + \frac{4}{13} \log_2 \frac{4}{13} \right) = 0.890$$

Right Group:

$$Yes = 0, \quad No = 1$$

$$Entropy(Right) = 0$$

Entropy for Threshold 630

Weighted Entropy:

$$WE = \frac{13}{14}(0.890) + \frac{1}{14}(0) = 0.827$$

Parent Entropy:

$$Entropy(S) = 0.940$$

Weighted Entropy:

$$WE = 0.827$$

Information Gain for Threshold 795

Information Gain:

$$IG = 0.940 - 0.827$$

$$IG = 0.113$$

This is the highest Information Gain among the tested CreditScore thresholds.

Step 7: Select the Root Node

Information Gain comparison:

$$IG(Age) = 0.247$$

$$IG(Income) = 0.029$$

$$IG(Student) = 0.152$$

$$IG(CreditScore) = 0.113$$

Highest Information Gain:

Age

Root Node = Age

Step 8: Split Dataset by Age

Root split:

Age

Age Group	IDs	Class Distribution	Decision
Young	1,2,8,9,11	2 Yes, 3 No	Not Pure
Middle	3,7,12,13	4 Yes, 0 No	Pure
Senior	4,5,6,10,14	3 Yes, 2 No	Not Pure

So:

Age = Middle $\rightarrow$ Buy = Yes

Young and Senior groups need further splitting.

Step 9: Young Group Calculation

After splitting by Age = Young:

ID	Income	Student	CreditScore	Buy
1	High	No	650	No
2	High	No	750	No
8	Medium	No	660	No
9	Low	Yes	640	Yes
11	Medium	Yes	730	Yes

Class distribution:

$Yes = 2, \quad No = 3$

$$Entropy(Young) = 0.971$$

Young Group: IG for Income

Split by Income:

Income	Yes	No	Entropy
High	0	2	0
Medium	1	1	1
Low	1	0	0

Weighted Entropy:

$$WE(\text{Income}) = \frac{2}{5}(0) + \frac{2}{5}(1) + \frac{1}{5}(0)$$

$$WE(\text{Income}) = 0.400$$

$$IG(\text{Income}) = 0.971 - 0.400$$

$$IG(\text{Income}) = 0.571$$

Young Group: IG for Student

Split by Student:

Student	Yes	No	Entropy
No	0	3	0
Yes	2	0	0

Weighted Entropy:

$$WE(Student) = \frac{3}{5}(0) + \frac{2}{5}(0)$$

$$WE(Student) = 0$$

$$IG(Student) = 0.971 - 0$$

$$IG(Student) = 0.971$$

Young Group: IG for CreditScore

Sorted CreditScore values:

640, 650, 660, 730, 750

Candidate thresholds:

645, 655, 695, 740

Best threshold:

$CreditScore \leq 645$

For this split:

Left : [Yes]

Right : [No, No, Yes, No]

$$Entropy(Left) = 0$$

$$Entropy(Right) = 0.811$$

Weighted Entropy:

$$WE(CreditScore) = \frac{1}{5}(0) + \frac{4}{5}(0.811)$$

$$WE(CreditScore) = 0.649$$

Information Gain:

$$IG(CreditScore) = 0.971 - 0.649$$

$$IG(CreditScore) = 0.322$$

Step 10: Best Split for Young Group

Information Gain inside Young group:

$$IG(\textit{Income}) = 0.571$$

$$IG(\textit{Student}) = 0.971$$

$$IG(\textit{CreditScore}) = 0.322$$

Highest Information Gain:

Student

Student = No $\rightarrow$ *Buy = No*

Student = Yes $\rightarrow$ *Buy = Yes*

Step 11: Senior Group Calculation

Senior group:

ID	Income	Student	CreditScore	Buy
4	Medium	No	680	Yes
5	Low	Yes	620	Yes
6	Low	Yes	800	No
10	Medium	Yes	710	Yes
14	Medium	No	760	No

Class distribution:

$$\text{Yes} = 3, \quad \text{No} = 2$$

$$\text{Entropy}(\text{Senior}) = 0.971$$

Senior Group: IG for Income

Split by Income:

Income	Yes	No	Entropy
Medium	2	1	0.918
Low	1	1	1

Weighted Entropy:

$$WE(Income) = \frac{3}{5}(0.918) + \frac{2}{5}(1)$$

$$WE(Income) = 0.951$$

$$IG(Income) = 0.971 - 0.951$$

$$IG(Income) = 0.020$$

Senior Group: IG for Student

Split by Student:

Student	Yes	No	Entropy
No	1	1	1
Yes	2	1	0.918

Weighted Entropy:

$$WE(Student) = \frac{2}{5}(1) + \frac{3}{5}(0.918)$$

$$WE(Student) = 0.951$$

$$IG(Student) = 0.971 - 0.951$$

$$IG(Student) = 0.020$$

Senior Group: IG for CreditScore

Sorted CreditScore values:

620, 680, 710, 760, 800

Candidate thresholds:

650, 695, 735, 780

Best threshold:

$CreditScore \leq 735$

Split:

Left : [Yes, Yes, Yes]

Right : [No, No]

Both groups are pure.

$$Entropy(Left) = 0$$

$$Entropy(Right) = 0$$

Weighted Entropy:

$$WE(CreditScore) = \frac{3}{5}(0) + \frac{2}{5}(0)$$

$$WE(CreditScore) = 0$$

Information Gain:

$$IG(CreditScore) = 0.971 - 0$$

$$IG(CreditScore) = 0.971$$

Step 12: Best Split for Senior Group

Information Gain comparison:

$$IG(\textit{Income}) = 0.020$$

$$IG(\textit{Student}) = 0.020$$

$$IG(\textit{CreditScore}) = 0.971$$

Highest Information Gain:

CreditScore

$$\textit{CreditScore} \leq 735 \rightarrow \textit{Buy} = \textit{Yes}$$

$$\textit{CreditScore} > 735 \rightarrow \textit{Buy} = \textit{No}$$

Final Decision Tree

Root Node: Age

Condition	Next Test	Prediction
Age = Middle	–	Yes
Age = Young	Student	Student=No $\rightarrow$ No, Student=Yes $\rightarrow$ Yes
Age = Senior	CreditScore	$\leq 735 \rightarrow$ Yes, $> 735 \rightarrow$ No

Final Rules Learned by the Tree

The Decision Tree learns the following rules:

If Age = Middle, then Buy Laptop = Yes

If Age = Young and Student = No, then Buy Laptop = No

If Age = Young and Student = Yes, then Buy Laptop = Yes

If Age = Senior and CreditScore $\leq$ 735, then Buy Laptop = Yes

If Age = Senior and CreditScore $>$ 735, then Buy Laptop = No

Tree Generation Summary

- 1 Calculate parent entropy.
- 2 Calculate Information Gain for all features.
- 3 Select feature with highest Information Gain.
- 4 Split the dataset.
- 5 If a child node is pure, make it a leaf node.
- 6 If a child node is not pure, repeat the process.
- 7 Continue until all nodes are pure or stopping criteria are reached.

Decision Tree in Python

```
from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier()

model.fit(X_train, y_train)

predictions = model.predict(X_test)
```

Scikit-Learn automatically:

- Calculates Entropy/Gini
- Finds best splits
- Builds the tree

Creating a Small Dataset

```
import pandas as pd

data = {
    'Age': [ 'Young', 'Young',
             'Young', 'Old',
             'Old', 'Old' ],

    'Income': [ 'High', 'Low',
                'High', 'High',
                'Low', 'Low' ],

    'Buy': [ 'No', 'No',
            'Yes', 'Yes',
            'Yes', 'Yes' ]
}
```

Visualizing the Tree

```
from sklearn.tree import plot_tree
import matplotlib.pyplot as plt

plot_tree(model,
          filled=True)

plt.show()
```

Visualization helps us understand the learned rules.

Important Hyperparameters

Parameter	Purpose
max_depth	Maximum tree depth
criterion	gini / entropy
min_samples_split	Minimum samples to split
min_samples_leaf	Minimum samples in leaf
max_features	Features considered

These control tree complexity.

What is Overfitting?

A tree can become too complex.

Symptoms:

- Very high training accuracy
- Poor test accuracy
- Memorizes training data

This is called Overfitting.

Overfitting Example

Small Tree:

- Generalizes well

Very Deep Tree:

- Creates too many branches
- Memorizes noise
- Poor performance on new data

Goal:

Balance bias and variance.

Preventing Overfitting

Common techniques:

- Limit max_depth
- Increase min_samples_split
- Increase min_samples_leaf
- Use Pruning
- Use Random Forest

These improve generalization.

Pre-Pruning:

- Stop growing early
- Restrict depth
- Restrict minimum samples

Post-Pruning:

- Grow full tree
- Remove unnecessary branches

Both reduce overfitting.

Applications of Decision Trees

- Medical Diagnosis
- Loan Approval
- Fraud Detection
- Customer Churn Prediction
- Employee Attrition Analysis
- Student Performance Prediction

Decision Trees are widely used because they are interpretable.

Summary

Today we learned:

- How Decision Trees are built
- Root node selection
- Recursive splitting
- Python implementation
- Hyperparameters
- Overfitting
- Pruning

Decision Trees form the foundation of Random Forest and Gradient Boosting.

Exercise

Consider the following dataset:

ID	Weather	Windy	Humidity	Play
1	Sunny	No	High	No
2	Sunny	Yes	High	No
3	Sunny	No	Normal	Yes
4	Rainy	No	Normal	Yes
5	Rainy	Yes	High	No
6	Rainy	No	High	Yes
7	Cloudy	No	Normal	Yes
8	Cloudy	Yes	Normal	Yes
9	Cloudy	Yes	High	No
10	Sunny	Yes	Normal	Yes

Tasks:

- 1 Draw the Decision Tree.

Thank You

Questions?